Ollscoil Mhá Nuad
Maynooth University

SEMESTER 1
2025-2026

CS310
Programming Languages & Compilers

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Time allowed: 2 hours

Answer at least ***three*** questions

Your mark will be based on your best ***three*** answers

All questions carry equal marks

Instructions

	Yes	No	N/A
Formulae and Tables book allowed *(i.e. available on request)*		X	
Formulae and Tables book required *(i.e. distributed prior to exam commencing)*		X	
Statistics Tables and Formulae allowed *(i.e. available on request)*		X	
Statistics Tables and Formulae required *(i.e. distributed prior to exam commencing)*		X	
Dictionary allowed *(supplied by the student)*		X	
Non-programmable calculator allowed		X	
Students required to write in and return the exam question paper	X		

Only the answers on this exam question paper will be marked – rough work here or on extra pages will not be marked.

Answer booklets will not be provided unless a student specifically requests one. If your answer requires more space than the exam question paper allows, at the bottom of your answer in the exam question paper, specify the page number(s) in your answer booklet where your answer continues.

Write your name, student number and desk number below.

Name ……..…….…….. Student no. ……..…….. Desk no.……

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[25 marks]

1 (a) Consider the regular expression: (a|b)*(ba|ab)(a|b)* over the alphabet Σ = {a, b}. Notice that the symbol '|' denotes 'or'. [4 marks]

Question. Which option best describes the language generated by this regular expression?

Options

i. All strings containing at least one 'a' and one 'b'.
ii. All strings containing exactly one 'b' and one 'a' in that order.
iii. All strings containing the substring 'ab' or 'ba'.
iv. All strings containing at least two 'b's and at least one 'a'.
v. All strings of length three.
vi. None of these.

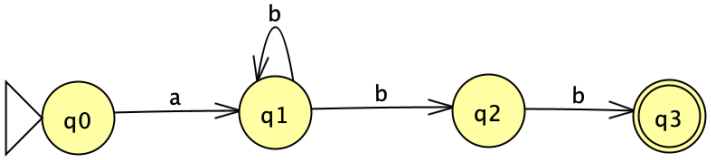
Start your answer like this:
The correct option is number __. All strings …
or
None of these.

Answer: ……...
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(b) Let *A* be the set containing the string "10" (one followed by zero). Define the set of all strings consisting of zero or more repetitions of "10" using operations on *A* such as union, concatenation, Kleene closure and/or positive closure. [4 marks]

Answer: ……...

(c) Given the following NFA (Σ = {a,b}) [6 marks]



Fill in the following transition table. Add or remove rows and/columns if necessary.

Rough work area: Transition Table

Answer correctly the following questions and use the correct notation as done during lectures to gain the marks.

i. What state(s) is or are reached from q0 on entry a?

Answer: ……...

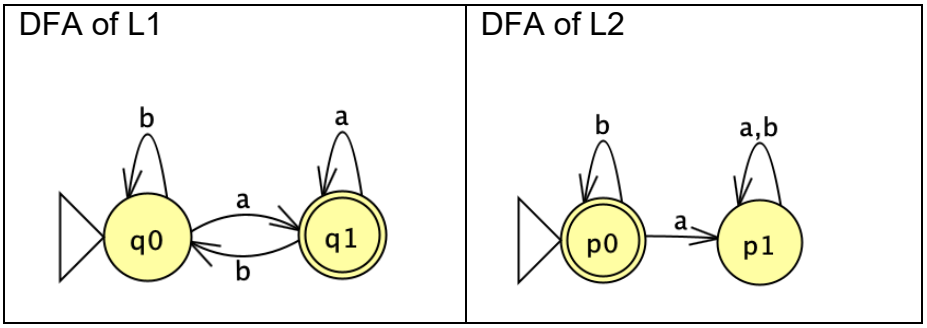
ii. What state(s) is or are reached from q3 on entry ε?

Answer: ……..

iii. What state(s) is or are reached from q2 on entry a?

Answer: ……...

(d) Construct the product automaton for L1 ∪ L2 and draw the corresponding Deterministic Finite Automaton (DFA) state diagram. Your diagram should use the same notation for states derived from combining the state sets of DFA L1 and DFA L2, for instance (q0,p0), (q1,p0), etc. The input alphabet is Σ={a,b} [8 marks]



Answer correctly the following questions on the resulting DFA to gain the marks:

i. What is or are the initial state(s)?

Answer: ……...

ii. What is or are the accepting state(s)?

Answer: ……...

iii. What state(s) is or are reached from the initial state(s) on entry a?

Answer: ……...

iv. What state(s) is or are reached from the initial state(s) on entry b?

Answer: ……...

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(e) Which of the following options best describes the language defined by the DFA of L1, as defined in 1(d), [3 marks]

Options

i. L={w|w is a string with an even number of 'a's }
ii. L={w|w is a string that contains at least one 'b' }
iii. L={w|w is a string that contains at least one 'a' }
iv. L={w|w is a string that starts with 'a' }
v. L={w|w is a string with an odd number of 'b's }
vi. None of these

Start your answer like this:
The correct option is number __. L={ …
or
None of these.

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Answer: ……...
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[25 marks]

2 (a) Let the alphabet Σ=Γ∪L, where Γ={α,β,γ} (three Greek letters) and L={a,b,c,d,e} (five Latin letters). Consider the language K=$(\Gamma \cup L)^2$. That is, all strings of length 2 over Σ. [8 marks]

Please note that cardinality of a set is the number of distinct elements in that set. Answer correctly the following questions to gain the marks.

i. What is the cardinality |K| (the number of length-2 strings over Σ)?

Answer: ……...

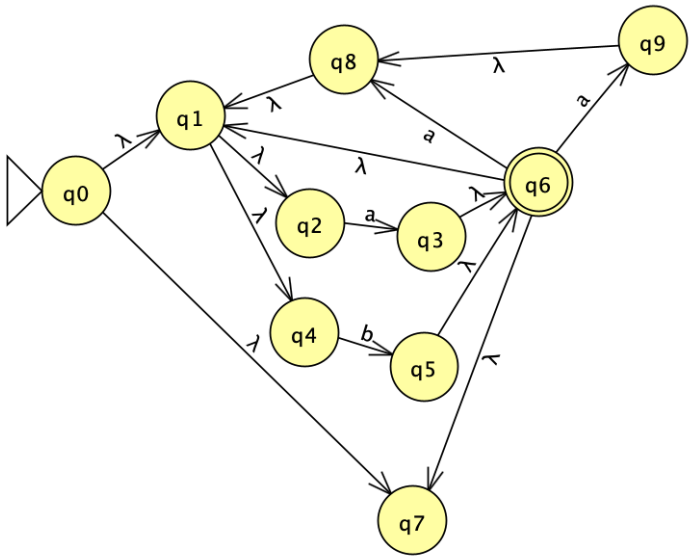
ii. What is the cardinality of the subset $K_{at_least_1_G} \subseteq K$ consisting of those length-2 strings that contain **at least one** symbol from Γ?

Answer: ……...

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(b) Convert the following NFA to DFA using the **subset construction algorithm**. Use A, B, C, etc to denote your resulting **ordered** states. Use Phi (Φ) if necessary. Σ = {a,b}. Notice that ⅄ is equivalent to epsilon, as used during lectures. For clarity purposes: [17 marks]

- Label a is used from q2 to q3, from q6 to q9, and from q6 to q8
- Label b is used from q4 to q5
- Label ⅄ is used for the remaining transitions denoted by arrows



Correctly answer the following questions based on your new DFA to gain the marks:

i. What is the number of resulting states?

Answer: ……...

ii. What state(s) is or are reached from the initial state(s) on entry **a**?

Answer: ……...

iii. What state(s) is or are reached from the initial state(s) on entry **b**?

Answer: ……...

iv. What is the number of final (accepting) states?

Answer: ……...

v. What state(s) is or are reached from State B on entry **a**?

Answer: ……...

vi. What state(s) is or are reached from Sate B on entry **b**?

Answer: ……...

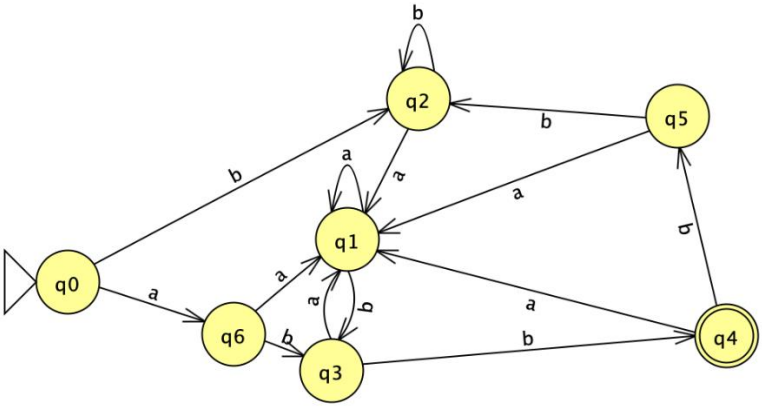
vii. How many Phi (Φ) states are in the new DFA?

Answer: ……...

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[25 marks]

3 (a) Minimise the following DFA using the state minimisation algorithm. **Use the same notation for the resulting states (e.g., q0, q1, q2, (q1,q2), etc) for the minimised DFA.** Σ = {a,b}. [10 marks]



Correctly answer the following questions based on the resulting DFA to gain the marks

i. What is the number of resulting states?

Answer: ……...

ii. What state(s) is or are reached from the initial state(s) on entry a?

Answer: ……...

iii. What state(s) is or are reached from the initial state(s) on entry b?

Answer: ……...

iv. What state(s) is or are reached from the accepting state(s) on entry a?

Answer: ……...

v. What state(s) is or are accepting states?

Answer: ……...

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(b) **Left factor** the following grammar. Clearly identify alpha (α) and beta (β) in the original grammar [15 marks]

$E \rightarrow i = E \mid i \bullet E$
$R \rightarrow id\ (list) \mid id\ [index] \mid id$
$B \rightarrow 0\ B\ 1 \mid 0\ 1$

Correctly answer the following questions to gain the marks.

i. What is the resulting grammar of E?

Answer: ……...

ii. What is the resulting grammar of E'?

Answer: ……...

iii. What is the resulting grammar of R?

Answer: ……...

iv. What is the resulting grammar of R'?

Answer: ……...

v. What is the resulting grammar of B?

Answer: ……...

vi. What is the resulting grammar of B'?

Answer: ……...

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[25 marks]

4 (a) Compute FIRST and FOLLOW for the following grammar, where ε represents epsilon. Fill in the following table to gain the marks: [25 marks]

Grammar	FIRST	FOLLOW
S → A B		
A → c D | f | ε		
B → g E B | d		
D → h | ε		
E → A j		

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